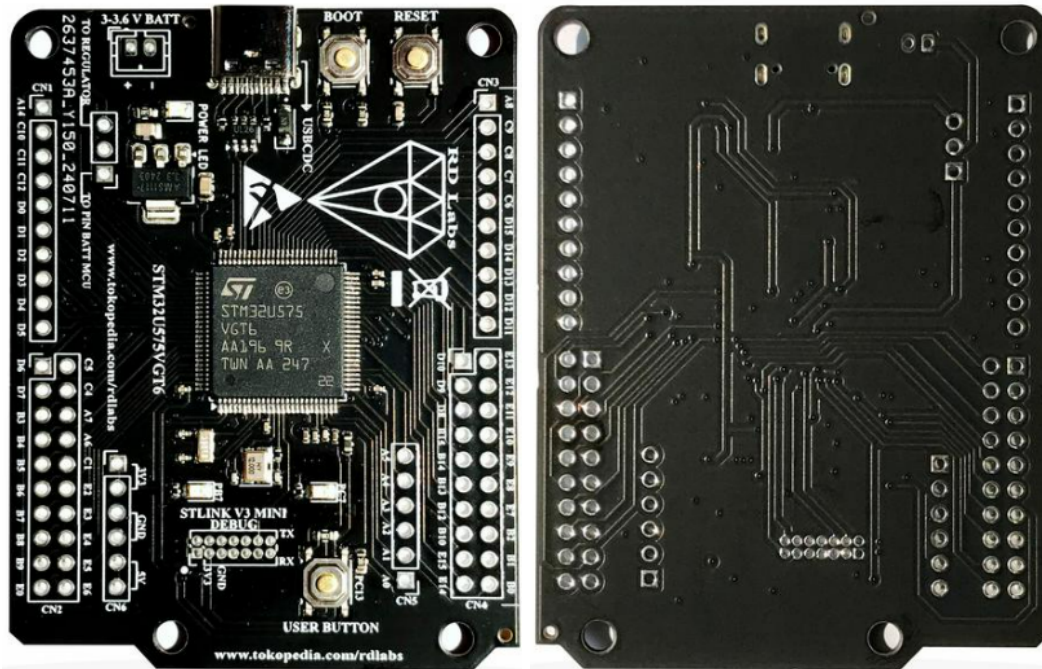




Description

The STM32U575VGT6 development board from RDLabs features the STM32U575VGT6 microcontroller, part of the STM32U5 ultra-low-power, high-performance family based on the Arm® Cortex®-M33 core, operating at frequencies up to 160 MHz. This core is equipped with a Floating Point Unit (FPU) for efficient floating-point operations, a full set of Digital Signal Processing (DSP) instructions, and a Memory Protection Unit (MPU) that enhances application security. The microcontroller includes large on-chip memory, with up to 2 MB of Flash and 786 KB of SRAM, making it suitable for data-intensive embedded applications.

The board offers advanced security features such as secure boot, secure firmware update, and secure firmware installation to protect code provisioning during production. Memory protection features include read, write, secure, and hidden area configurations for both Flash and SRAM. The device also supports anti-tamper detection and protection against environmental or physical attacks. Analog peripherals include high-speed 14-bit and 12-bit ADCs (2.5 Msps), two DAC channels, internal op-amps and comparators, and a reference voltage buffer. It also features dedicated hardware accelerators such as the Chrom-ART graphics accelerator, trigonometric and



filter math accelerators, an Audio Digital Filter (ADF), and a Multi-function Digital Filter (MDF) for interfacing with external sigma-delta sensors.

On the communication side, the board supports various standard and advanced interfaces including 4x I2C, 3x SPI, 3x USART, 2x UART, 1x low-power UART, 2x SAI, 2x SDMMC, USB OTG Full-Speed (available via the onboard USB-C port), a USB Type-C/Power Delivery controller, FDCAN, DCMI, and PSSI. It also includes up to 22 capacitive touch sensing channels. The board operates from a 1.71 V to 3.6 V supply and supports an industrial temperature range from -40°C to +125°C. Power-saving features such as Stop Mode and Low-Power Background Autonomous Mode (LPBAM) make it ideal for battery-powered and IoT applications. Additionally, the board includes a VBAT input for preserving RTC and backup SRAM content during power loss. Overall, this board is an excellent choice for secure, low-power embedded applications involving sensors, communication protocols, signal processing, and real-time data handling.

Features

Key Features of RDLabs STM32U575VGT6 Development Board Ultra-Low Power and Power Management

- Operates from 1.71 V to 3.6 V supply voltage
- Supports Low-Power Background Autonomous Mode (LPBAM)
- Multiple low-power modes: Shutdown (160 nA), Standby with RTC (530 nA), Stop 2 (as low as 4 µA)
- VBAT support for RTC, backup SRAM (2 KB), and 32 × 32-bit backup registers

Core and Performance

- Arm® Cortex®-M33 CPU with TrustZone®, FPU, MPU, and DSP
- Runs up to 160 MHz (240 DMIPS, 651 CoreMark®)
- 8 KB instruction cache and 4 KB data cache for fast memory access
- Integrated LDO and SMPS regulators for dynamic voltage scaling



Memory

- 2 MB Flash with ECC (Read-While-Write support)
- 786 KB SRAM (including ECC-enabled regions)
- Support for external memories via Octo-SPI interfaces

Security Features

- Arm® TrustZone® support for secure application isolation
- Secure Boot, Secure Firmware Install (SFI), and Secure Firmware Update\
- Hardware cryptographic support: HASH accelerator, True Random Number Generator
- Active tamper detection and debug lock with password

Clock and Timers

- Internal/external oscillators: 16 MHz HSI, 48 MHz, and 32 kHz LSE for RTC
- Up to 17 timers, including advanced motor-control and low-power timers
- RTC with calendar and calibration support

Analog and Digital Peripherals

- 14-bit and 12-bit ADCs with up to 2.5 Msps sampling
- 2 DACs, 2 Operational Amplifiers, 2 Comparators
- CRC calculation unit

Connectivity

- USB Type-C / USB OTG FS via onboard USB-C connector
- 4x I2C, 3x SPI, 3x USART, 2x UART, 1x Low-Power UART
- 2x SAI, 2x SDMMC, CAN FD, PSSI (Parallel Slave Interface)



- Support for capacitive touch sensing (up to 22 channels)

Graphics and Math Acceleration

- Chrom-ART Accelerator (DMA2D) for graphics
- CORDIC for trigonometric acceleration and FMAC for signal processing

Development and Debug

- Supported by STLINK V3 Mini debugger (available on board)
- Debug via SWD, JTAG, and Embedded Trace Macrocell (ETM)
- User-accessible BOOT and RESET buttons, as well as User Button



Board Connectors

Connector	Pin Number	Pin Name	Function
CN1	1	PA15	JTDI, TIM2_CH1, TIM2_ETR, USART2_RX, SPI1_NSS, SPI3_NSS, USART3_RTS_DE, UART4_RTS_DE, SAI2_FS_B, EVENTOUT
	2	PC10	TRACED1, LPTIM3_ETR, ADF1_CCK1, SPI3_SCK, USART3_TX, UART4_TX, TSC_G3_IO2, DCMI_D8, LPGPIO1_P8, SDMMC1_D2, SAI2_SCK_B, EVENTOUT
	3	PC11	LPTIM3_IN1, ADF1_SDI0, DCMI_D2, OCTOSPIM_P1_NCS, SPI3_MISO, USART3_RX, UART4_RX, TSC_G3_IO3, DCMI_D4, UCPD1_FRSTX2, SDMMC1_D3, SAI2_MCLK_B, EVENTOUT
	4	PC12	TRACED3, SPI3_MOSI, USART3_CK, UART5_TX, TSC_G3_IO4, DCMI_D9, LPGPIO1_P10, SDMMC1_CK, SAI2_SD_B, EVENTOUT
	5	PD0	TIM8_CH4N, SPI2_NSS, FDCAN1_RX, FMC_D2, EVENTOUT
	6	PD1	SPI2_SCK, FDCAN1_TX, FMC_D3, EVENTOUT
	7	PD2	TRACED2, TIM3_ETR, USART3_RTS_DE, UART5_RX, TSC_SYNC, DCMI_D11, LPGPIO1_P7, SDMMC1_CMD, LPTIM4_ETR, EVENTOUT
	8	PD3	SPI2_SCK, DCMI_D5, SPI2_MISO, MDF1_SDI0, USART2_CTS, OCTOSPIM_P2_NCS, FMC_CLK, EVENTOUT
	9	PD4	SPI2_MOSI, MDF1_CK10, USART2_RTS_DE, OCTOSPIM_P1_IO4, FMC_NOE, EVENTOUT
	10	PD5	SPI2_RDY, USART2_TX, OCTOSPIM_P1_IO5, FMC_NWE, EVENTOUT

Connector	Pin Number	Pin Name	Function
	1	PD6	SAI1_D1, DCMI_D10, PSSI_D10, SPI3_MOSI, MDF1_SDI1, USART2_RX, OCTOSPIM_P1_IO6, SDMMC2_CK, FMC_NWAIT, SAI1_SD_A, EVENTOUT
	2	PC5	TIM1_CH4N, SAI1_D3, PSSI_D15, USART3_RX, EVENTOUT, COMP1_INP1, ADC1_IN14, ADC4_IN23, WKUP5, TAMP_IN4/OUT5
	3	PD7	MDF1_CK11, USART2_CK, OCTOSPIM_P1_IO7,



CN2			SDMMC2_CMD, FMC_NCE/FMC_NE1, LPTIM4_OUT, EVENTOUT
	4	PC4	USART3_TX, OCTOSPIM_P1_IO7, EVENTOUT, COMP1_INM2, ADC1_IN13, ADC4_IN22
	5	PB3	JTDO/TRACESWO, TIM2_CH2, LPTIM1_CH1, ADF1_CCK0, I2C1_SDA, SPI1_SCK, SPI3_SCK, USART1_RTS_DE, CRS_SYNC, LPGPIO1_P11, SDMMC2_D2, SAI1_SCK_B, EVENTOUT
	6	PA7	SRDSTOP, TIM1_CH1N, TIM3_CH2, TIM8_CH1N, I2C3_SCL, SPI1_MOSI, USART3_TX, OCTOSPIM_P1_IO2, LPTIM2_CH2, TIM17_CH1, EVENTOUT, ADC1_IN12, ADC4_IN20
	7	PB4	NJTRST, LPTIM1_CH2, TIM3_CH1, ADF1_SDIO, I2C3_SDA, SPI1_MISO, SPI3_MISO, USART1_CTS, UART5_RTS_DE, TSC_G2_IO1, DCM1_D12, SDMMC2_D3, SAI1_MCLK_B, TIM17_BKIN, EVENTOUT
	8	PA6	CDSTOP, TIM1_BKIN, TIM3_CH1, TIM8_BKIN, DCM1_PIXCLK, SPI1_MISO, USART3_CTS, LPUART1_CTS, OCTOSPIM_P1_IO3, TIM16_CH1, EVENTOUT, ADC1_IN11, ADC4_IN11
	9	PB5	LPTIM1_IN1, TIM3_CH2, OCTOSPIM_P1_NCLK, I2C1_SMBA, SPI1_MOSI, SPI3_MOSI, USART1_CK, UART5_CTS, TSC_G2_IO2, DCM1_D10, SAI1_SD_B, TIM16_BKIN, EVENTOUT
	10	PA5	CSLEEP, TIM2_CH1, TIM2_ETR, TIM8_CH1N, PSSI_D14, SPI1_SCK, USART3_RX, LPTIM2_ETR, EVENTOUT, ADC1_IN10, ADC4_IN10
	11	PB6	LPTIM1_ETR, TIM4_CH1, TIM8_BKIN2, I2C1_SCL, I2C4_SCL, MDF1_SDI5, USART1_TX, TSC_G2_IO3, DCM1_D5, SAI1_FS_B, TIM16_CH1N, EVENTOUT
	12	PE2	TRACED2, DCM1_D4, FMC_A23, EVENTOUT
	13	PB7	LPTIM1_IN2, TIM4_CH2, TIM8_BKIN, I2C1_SDA, I2C4_SDA, MDF1_CK15, USART1_RX, UART4_CTS, DCM1_VSYNC, TIM17_CH1N, EVENTOUT, LED
	14	PE3	DCM1_D5, FMC_A19, EVENTOUT
	15	PB8	TIM4_CH3, SAI1_CK1, I2C1_SCL, MDF1_CCK0, SPI3_RDY, SDMMC1_CKIN, FDCAN1_RX, DCM1_D6, TIM16_CH1, EVENTOUT
	16	PE4	DCM1_D6, FMC_A20, EVENTOUT
	17	PB9	IR_OUT, TIM4_CH4, SAI1_D2, I2C1_SDA, SPI2_NSS, SDMMC1_CD1R, FDCAN1_TX, DCM1_D7, TIM17_CH1,



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			EVENTOUT
	18	PE5	DCMI_D7, FMC_A21, EVENTOUT
	19	PE0	TIM4_ETR, DCMI_D2, FMC_NBL0, TIM16_CH1, EVENTOUT
	20	PE6	DCMI_D3, FMC_NBL1, TIM17_CH1, EVENTOUT

Connector	Pin Number	Pin Name	Function
CN3	1	PA8	MCO, TIM1_CH1, SAI1_CK2, SPI1_RDY, USART1_CK, OTG_FS_SOF, TRACECLK, SAI1_SCK_A, LPTIM2_CH1, EVENTOUT
	2	PC9	TRACED0, TIM8_BKIN2, TIM3_CH4, TIM8_CH4, DCMI_D3/PSSI_D3, TSC_G4_IO4, OTG_FS_NOE, SDMMC1_D1, LPTIM3_CH2, EVENTOUT
	3	PC8	SRDSTOP, TIM3_CH3, TIM8_CH3, TSC_G4_IO3, DCMI_D2/PSSI_D2, SDMMC1_D0, LPTIM3_CH1, EVENTOUT
	4	PC7	CDSTOP, TIM3_CH2, TIM8_CH2, MDF1_SDI3, SDMMC1_D123DIR, TSC_G4_IO2, DCMI_D1/PSSI_D1, SDMMC2_D7, SDMMC1_D7, SAI2_MCLK_B, LPTIM2_CH2, EVENTOUT, LED
	5	PC6	CDSTOP, TIM3_CH2, TIM8_CH2, MDF1_SDI3, SDMMC1_D123DIR, TSC_G4_IO2, DCMI_D1/PSSI_D1, SDMMC2_D7, SDMMC1_D7, SAI2_MCLK_B, LPTIM2_CH2, EVENTOUT
	6	PD15	TIM4_CH4, FMC_D1, EVENTOUT
	7	PD14	TIM4_CH3, FMC_D0, EVENTOUT
	8	PD13	TIM4_CH2, FMC_A18, EVENTOUT
	9	PD12	TIM4_CH2, FMC_A18, EVENTOUT
	10	PD11	USART3_CTS, CAN1_RX, TSC_G6_IO3, EVENTOUT

Connector	Pin Number	Pin Name	Function
	1	PD10	LPTIM2_CH2, USART3_CK, TSC_G6_IO1, FMC_D15, SAI2_SCK_A, LPTIM3_ETR, EVENTOUT
	2	PE13	TIM1_CH3, SPI1_SCK, MDF1_CK15, TSC_G5_IO4, OCTOSPIM_P1_IO1, FMC_D10, EVENTOUT



CN4	3	PD9	LPTIM2_IN2, USART3_RX, DCMI_PIXCLK/PSSI_PDCK, FMC_D14, SAI2_MCLK_A, LPTIM3_IN1, EVENTOUT
	4	PE12	TIM1_CH3N, SPI1_NSS, MDF1_SDI5, TSC_G5_IO3, OCTOSPIM_P1_IO0, FMC_D9, EVENTOUT
	5	PD8	USART3_TX, DCMI_HSYNC/PSSI_DE, FMC_D13, EVENTOUT
	6	PE11	TIM1_CH2, SPI1_RDY, MDF1_CK14, TSC_G5_IO2, OCTOSPIM_P1_NCS, FMC_D8, EVENTOUT
	7	PB15	RTC_REFIN, TIM1_CH3N, LPTIM2_IN2, TIM8_CH3N, SPI2_MOSI(boot), MDF1_CK12, FMC_NBL1, SDMMC2_D1, SAI2_SD_A, TIM15_CH2, EVENTOUT
	8	PE10	TIM1_CH2N, ADF1_SDI0, MDF1_SDI4, TSC_G5_IO1, OCTOSPIM_P1_CLK, FMC_D7, SAI1_MCLK_B, EVENTOUT
	9	PB14	TIM1_CH2N, LPTIM3_ETR, TIM8_CH2N, I2C2_SDA, SPI2_MISO(boot), MDF1_SDI2, USART3_RTS_DE, TSC_G1_IO3, SDMMC2_D0, SAI2_MCLK_A, TIM15_CH1, EVENTOUT
	10	PE9	TIM1_CH1, ADF1_CCK0, MDF1_CCK0, OCTOSPIM_P1_NCLK, FMC_D6, SAI1_FS_B, EVENTOUT
	11	PB13	TIM1_CH1N, LPTIM3_IN1, I2C2_SCL, SPI2_SCK, MDF1_CK11, USART3_CTS, LPUART1_CTS, TSC_G1_IO2, SAI2_SCK_A, TIM15_CH1N, EVENTOUT
	12	PE8	TIM1_CH1N, MDF1_CK12, FMC_D5, SAI1_SCK_B, EVENTOUT, WKUP7
	13	PB12	TIM1_BKIN, I2C2_SMBA, SPI2_NSS, MDF1_SDI1, USART3_CK, LPUART1_RTS_DE, TSC_G1_IO1, OCTOSPIM_P1_NCLK, SAI2_FS_A, TIM15_BKIN, EVENTOUT
	14	PE7	TIM1_ETR, MDF1_SDI2, FMC_D4, SAI1_SD_B, EVENTOUT, WKUP6
	15	PB11	TIM2_CH4, I2C4_SDA, I2C2_SDA, SPI2_RDY, USART3_RX, LPUART1_TX, OCTOSPIM_P1_NCS, COMP2_OUT, EVENTOUT
	16	PE1	OCTOSPIM_P1_IO1, FMC_NBL1, EVENTOUT
	17	PB10	TIM2_CH3, LPTIM3_CH1, I2C4_SCL, I2C2_SCL, SPI2_SCK, USART3_TX, LPUART1_RX, TSC_SYNC, OCTOSPIM_P1_CLK, COMP1_OUT, SAI1_SCK_A, EVENTOUT
	18	PE0	OCTOSPIM_P1_IO0, FMC_NBL0, TIM16_CH1, EVENTOUT



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	19	PE14	TSC_G6_IO4, OCTOSPIM_P2_DQS, TIM1_CH4, FMC_D11, EVENTOUT
	20	PB0	TIM1_CH2N, TIM3_CH3, TIM8_CH2N, LPTIM3_CH1, SPI1_NSS, USART3_CK, OCTOSPIM_P1_IO1, LPGPIO1_P9, COMP1_OUT, AUDIOCLK, OPAMP2_VOUT, ADC1_IN15, ADC4_IN18, EVENTOUT

Connector	Pin Number	Pin Name	Function
CN5	1	PA0	TIM2_CH1, TIM5_CH1, TIM8_ETR, LPTIM1_IN1, COMP1_INP, OPAMP1_VINP, ADC1_IN5, ADC4_IN5, WKUP1, EVENTOUT
	2	PA1	TIM2_CH2, TIM5_CH2, LPTIM1_OUT, COMP1_INM, OPAMP1_VINM, ADC1_IN6, ADC4_IN6, WKUP2, EVENTOUT
	3	PA2	TIM2_CH3, TIM5_CH3, LPTIM1_ETR, COMP2_INP, OPAMP2_VINP, ADC1_IN7, ADC4_IN7, USART2_TX, EVENTOUTGPIO, ADC1_IN7, USART2_TX
	4	PA3	TIM2_CH4, TIM5_CH4, LPTIM1_IN2, COMP2_INM, OPAMP2_VINM, ADC1_IN8, ADC4_IN8, USART2_RX, EVENTOUT
	5	PA4	SPI1_NSS, SPI3_NSS, USART2_CK, DCMI_HSYNC, SAI1_FS_B, LPTIM2_CH1, ADC1_IN9, ADC4_IN9, DAC1_OUT1, WKUP2, EVENTOUT
	6	PA5	TIM2_CH1, TIM2_ETR, TIM8_CH1N, PSSI_D14, SPI1_SCK, USART3_RX, LPTIM2_ETR, ADC1_IN10, ADC4_IN10, DAC1_OUT2, WKUP6, EVENTOUT

Connector	Pin Number	Pin Name	Function
CN6	1	3.3V	POWER SUPPLY
	2	3.3V	POWER SUPPLY
	3	GND	GROUND
	4	GND	GROUND
	5	5V	USB POWER INPUT
	6	5V	USB POWER INPUT

Connector	Pin	Pin Name	Function
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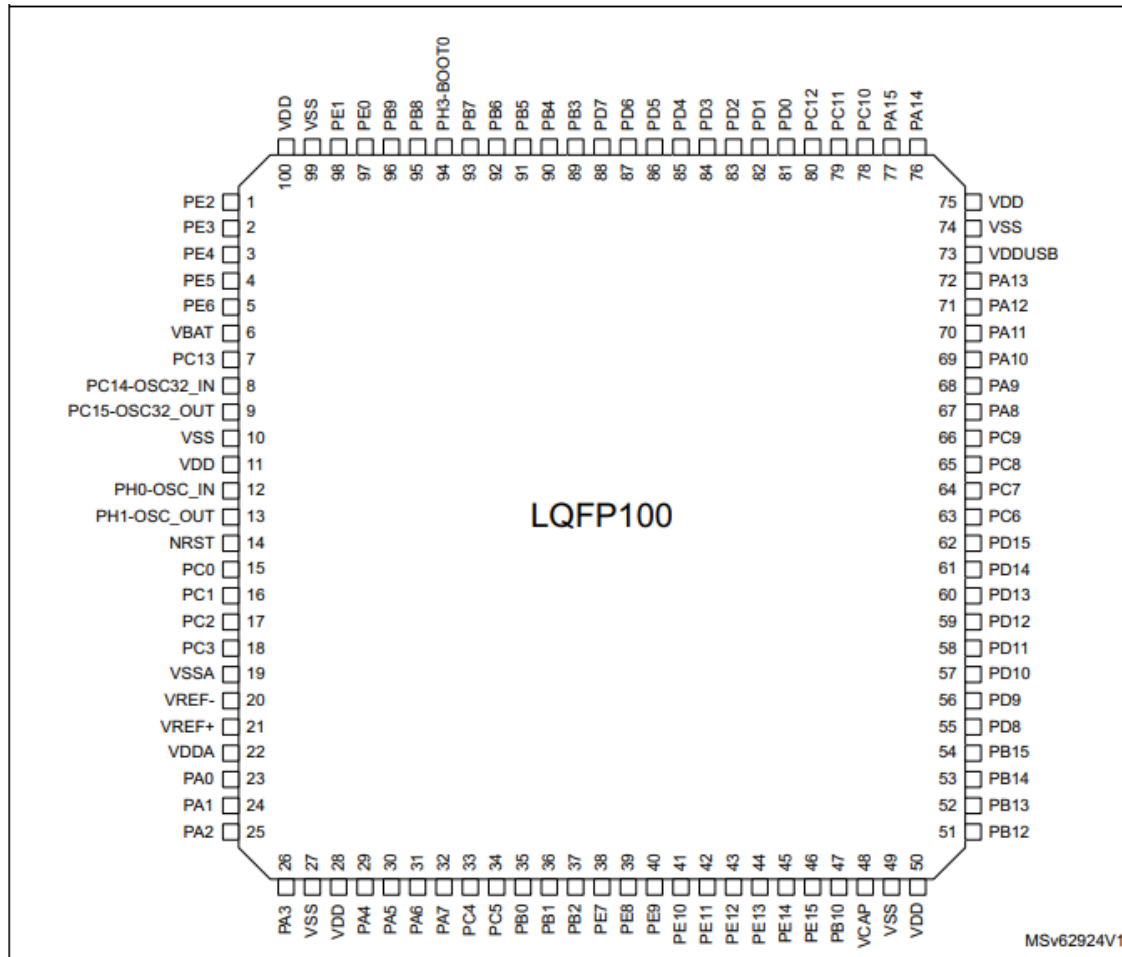


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	Number		
STLINK V3MINI	1	-	
	2	-	
	3	3.3V	POWER SUPPLY
	4	SWDIO	SERIAL WIRE DEBUG I/O
	5	GND	GROUND
	6	SWCLK	SERIAL WIRE CLOCK
	7	GND	GROUND
	8	-	
	9	-	
	10	-	
	11	GND	GROUND
	12	NRST	RESET
	13	USART1_RX	USART1_RX
	14	USART1_TX	USART1_TX



Pin Out





Getting Started with STM32U575VGT6

Requirements

- STM32U575VGT6 board (RDlabs)
- STLink-V3MINI programmer/debugger
- STM32CubeIDE (latest version)
- USB type-C cable
- 1x LED connected to PC7 onboard

Step by step

1. Open or install **STM32CubeIDE**.
2. Go to **File** → **New** → **STM32 Project**.
3. In the **Board Selector** or **MCU Selector**, type STM32U575VGT6 and click **Next**.
4. Name your project ex: LED_Blink and click **Finish**.
5. In the **Pinout & Configuration** tab, Click on **PC7** pin and set it to **GPIO_Output**.
6. Go to **Project Manager** → **Advanced Settings**: Under **Driver**, set **GPIO** to **HAL**.
7. Click **Save** (the code will be generated).
8. Open Src/main.c, and inside the while(1) loop, add the blinking logic:

Program:

```
/* USER CODE BEGIN WHILE */

while (1)

{

    HAL_GPIO_TogglePin(GPIOC, GPIO_PIN_7);

    HAL_Delay(500); // 500 ms delay

}

/* USER CODE END WHILE */
```



9. Make sure PC7 is initialized correctly in MX_GPIO_Init() function (should be auto-generated):

Function example:

```
GPIO_InitStruct.Pin = GPIO_PIN_7;  
  
GPIO_InitStruct.Mode = GPIO_MODE_OUTPUT_PP;  
  
GPIO_InitStruct.Pull = GPIO_NOPULL;  
  
GPIO_InitStruct.Speed = GPIO_SPEED_FREQ_LOW;  
  
HAL_GPIO_Init(GPIOC, &GPIO_InitStruct);
```

10. Click **"Build"** (hammer icon) to compile the project.
11. Connect the **ST-Link V3MINI** to your PC and target board.
12. Click the **green bug icon ("Debug")** or the **"Run"** button (green triangle).
13. If the setup is correct, you should see the LED on **PC7** blinking every 500ms.